

Trainings ▾

General Information

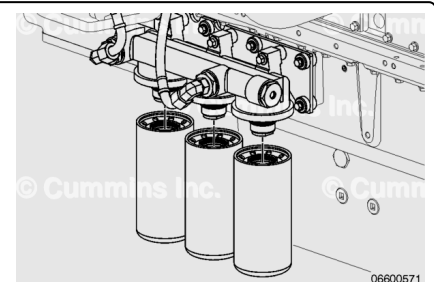
Note : This procedure applies to Stage 2 fuel filters on engines with electronically actuated injectors only.

- Use the following procedure for engines with mechanically actuated injectors. Refer to Procedure 006-015 in Section 4.
(/qs3/pubsys2/xml/en/procedures/28/28-006-015-om.html)
- Use the following procedure for stage 1 fuel filters. Refer to Procedure 006-075 in Section 4.
(/qs3/pubsys2/xml/en/procedures/28/28-006-075-om.html)



All Applications

The Stage 2 filter head is mounted on the engine and is vibration-isolated. There is an option for the Stage 2 filter to be remote-mounted, depending on the OEM.

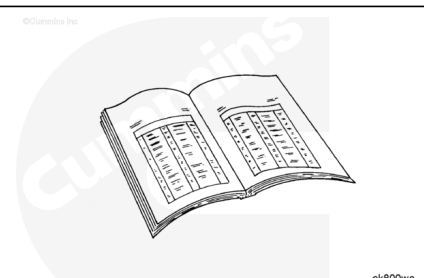


Preparatory Steps

with Electronically Actuated Injector

⚠ WARNING ⚠

Depending on the circumstance, fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe



personal injury, death or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.

Industrial and Power Generation Applications

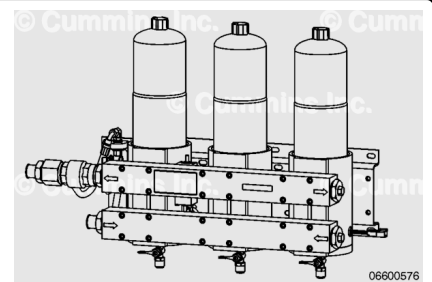
- The Stage 2 filtration consists of an engine-mounted, vibration-isolated dual filter head. The Stage 2 filtration **must** be used during normal engine operation.

Shut down the engine.

Close the fuel supply shutoff valve.

⚠ WARNING ⚠

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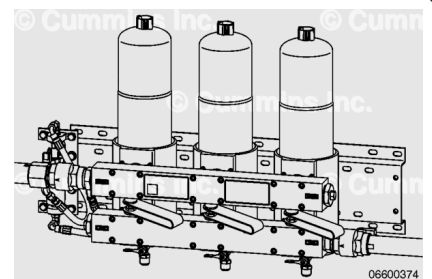


Marine Dual/Triple filter Applications

- The Stage 2 filtration consists of a vibration-isolated triple filter head that is engine-mounted. The Stage 2 filtration **must** be used during engine operation.

⚠ WARNING ⚠

Depending on the circumstance, fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.



Note : This procedure explains Triplex fuel system filter changes with the engine operating. If the engine is Triplex/Quadplex, and the intent is to change filters with the engine shut off, follow the Dual/Triple filter procedures.

Marine Triplex/Quadplex Applications

- The Stage 2 filtration consists of an engine-mounted vibration-isolated filter head. All three filter elements **must** be used during engine operation.

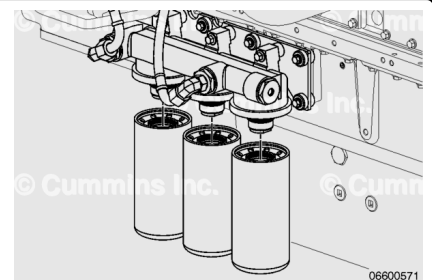
The Stage 2 filters can **not** be changed while the engine is under operation. Replace filters **only** when the engine is shut off.

Remove

with Electronically Actuated Injector

⚠ WARNING ⚠

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Remove the fuel filters. Use fuel filter wrench, Part Number 3400158, or equivalent.

Clean

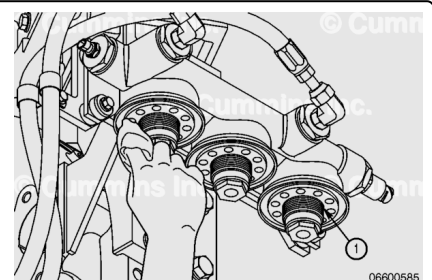
with Electronically Actuated Injector

Check that the thread adapter sealing ring (1) has been removed.

Use a clean lint-free cloth to clean the gasket mating surface of the filter head.

Spray the gasket mating surface of the filter head with cleaning solvent.

Allow the gasket mating surface of the filter head to air-dry.



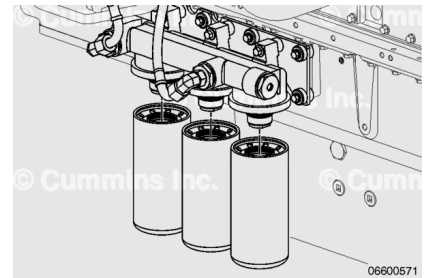
Install

with Electronically Actuated Injector

Note : Do not fill the Stage 2 filters with fuel.

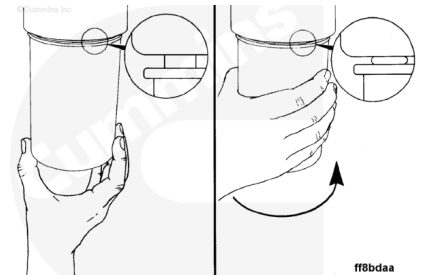
Lubricate the filter o-rings with clean engine oil.

Install the Stage 2 fuel filters on the triple filter head.



Turn the filter until the o-ring touches the surface of the filter head.

Tighten the filter an additional $\frac{1}{2}$ to $\frac{3}{4}$ of a turn after the o-ring touches the filter head surface.



Prime

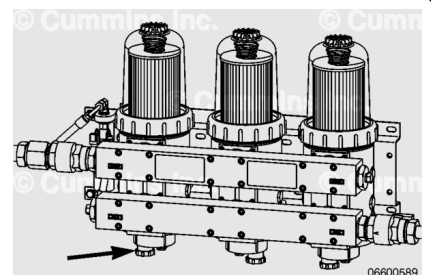
with Electronically Actuated Injector

Industrial and Power Generation Applications

- Stage 1 and Stage 2 filters **must** be primed prior to starting the engine.
- The procedure for priming Stage 1 and Stage 2 filtration is covered in the same steps, as follows:

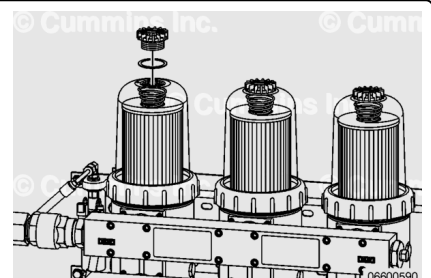
Verify the drain valve on the Stage 1 filter is closed at the base of the filter.

Close the fuel supply shutoff valve.



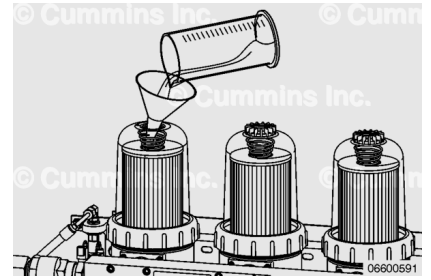
If the stage 1 filters have drained, they must be filled prior to priming the stage 2 filters.

Remove the vent cap from the top of the filter.



⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when filling the Stage 1 filter. Use clean fuel, a clean container, and a clean funnel when filling the Stage 1 filter. Use of clean components is necessary to prevent any dirt or foreign particles from entering the fuel system. Dirt or foreign particles can cause damage to the fuel system.



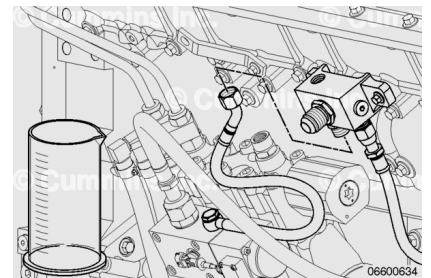
Fill the filter with clean diesel fuel.

Install the vent cap.

Tighten the vent cap by hand **only**.

Open the fuel supply shutoff valve.

Remove the air bleed line from the drain manifold block and place it into a collection container.

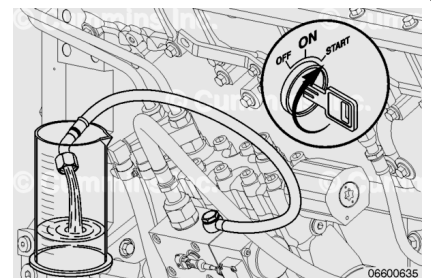


Turn the keyswitch ON and observe fuel exiting the air bleed line.

When a steady stream of fuel is observed, turn the keyswitch OFF.

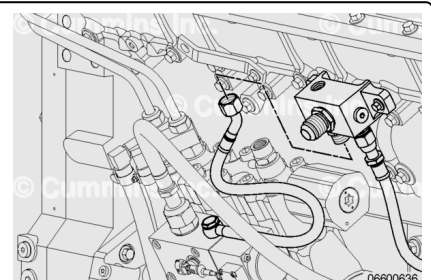
If the priming pump stops operating before a steady stream of fuel is observed, turn the keyswitch OFF, wait 30 seconds, and turn the keyswitch ON.

Repeat as necessary to observe a steady stream of fuel.



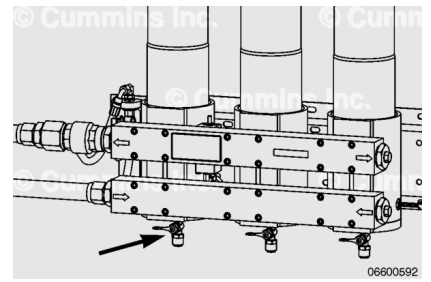
Install the air bleed line into the drain manifold block.

Torque Value: 45 n·m [33 ft·lb]



⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when filling the Stage 1 filter. Use clean fuel, a clean container, and a clean funnel when filling the Stage 1 filter. Use of clean components is necessary to prevent any dirt or foreign particles from entering the fuel system. Dirt or foreign particles can cause damage to the fuel system.



Marine Dual/Triple Filter Applications

- Stage 1 and Stage 2 filters **must** be primed prior to starting the engine.
- The procedure for priming Stage 1 and Stage 2 filtration is covered in the same steps, as follows.

Stage 1 Priming:

Verify the drain valve is closed at the base of the Stage 1 filter.

Close the fuel supply shutoff valve.

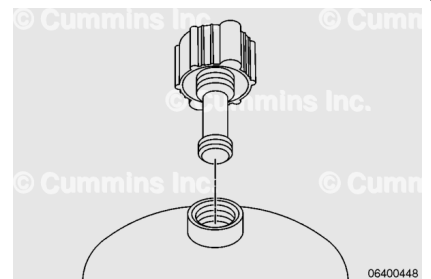
If the Stage 1 filters have been drained, they **must** be filled prior to priming the Stage 2 filters.

Unscrew the threads of the vent cap.

Lift the vent cap until the retainer shaft catches.

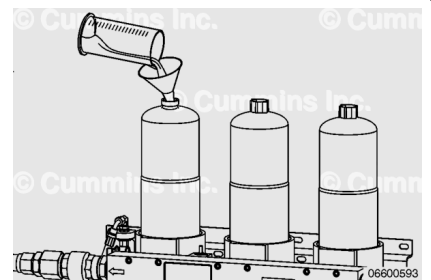
Unscrew the retainer shaft from the threaded boss in the top of the filter can.

Remove the vent cap from the top of the can.



⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when filling the Stage 1 filter. Use clean fuel, a clean container, and a clean funnel when filling the Stage 1 filter. Use of clean components is necessary to prevent any dirt or foreign particles from entering the fuel system. Dirt or foreign particles can cause damage to the fuel system.



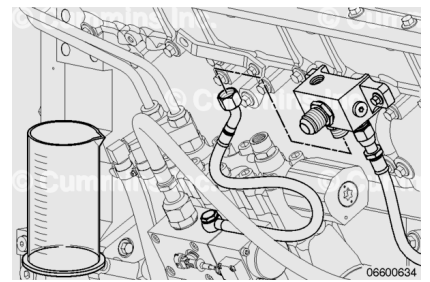
Fill the filter with clean diesel fuel.

Install the vent cap.

Tighten the vent cap by hand **only**.

Open the fuel supply shutoff valve.

Remove the air bleed line from the drain manifold block and place it into a collection container.

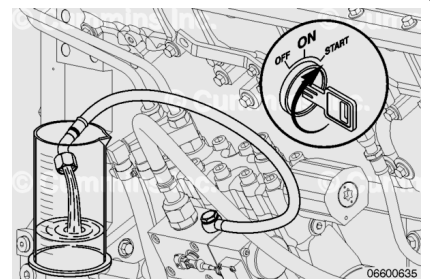


Turn the keyswitch ON and observe fuel exiting the air bleed line.

When a steady stream of fuel is observed, turn the keyswitch OFF.

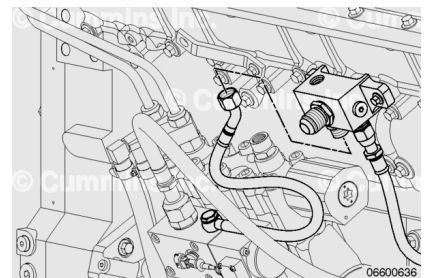
If the priming pump stops operating before a steady stream of fuel is observed, turn the keyswitch OFF, wait 30 seconds, and turn the keyswitch ON.

Repeat as necessary to observe a steady stream of fuel.



Install the air bleed line into the drain manifold block.

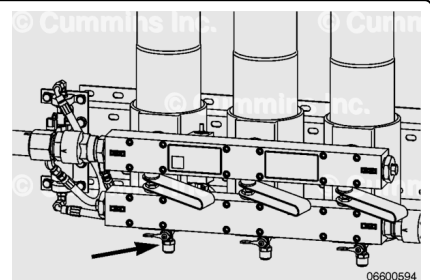
Torque Value: 45 n•m [33 ft-lb]



Note : This procedure explains Triplex/Quadplex fuel system filter changes with the engine operating at idle. If the engine is Triplex/Quadplex, and the intent is to change filters with the engine shutoff, follow the Dual/Triple filter procedures.

Marine Triplex/Quadplex Applications

- Stage 1 filters **must** be primed prior to resuming filter operation.
- The procedure for priming Stage 1 filtration is as follows:



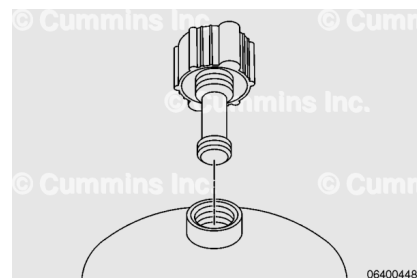
If the Stage 1 filters have been drained, they **must** be filled prior to priming the Stage 2 filters.

Unscrew the threads of the vent cap.

Lift the vent cap until the retainer shaft catches.

Unscrew the retainer shaft from the threaded boss in the top of the filter can.

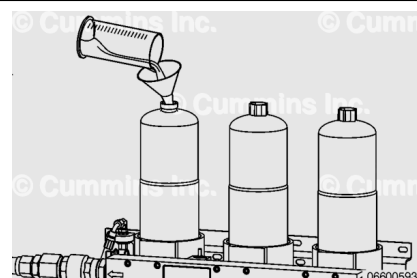
Remove the vent cap from the top of the can.



Fill the filter with clean diesel fuel.

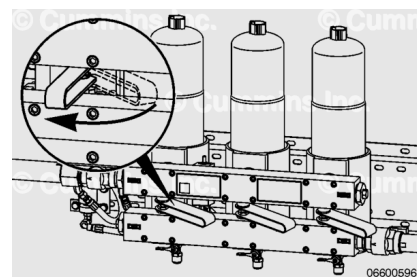
Install the vent cap.

Tighten the vent cap by hand **only**.



Slowly open the valve handle to the ON position, allowing excess time to purge air from the system.

Repeat the procedure for the remaining filter replacements.

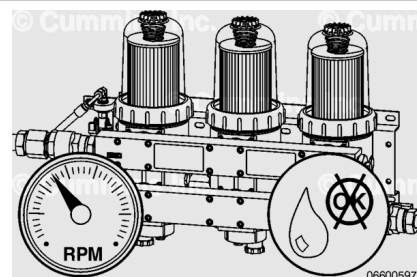


Finishing Steps

with Electronically Actuated Injector

⚠ WARNING ⚠

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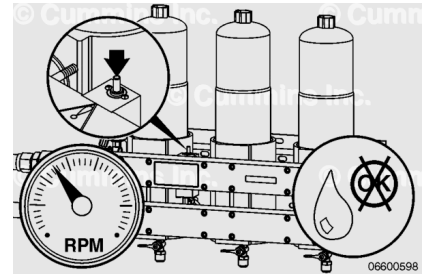
Industrial and Power Generation Applications

- The Industrial Pro™ filter **must** be checked for leaks.
- Check the plumbing to and from the Stage 1 and Stage 2 filter heads.

- Operate the engine for 1 to 2 minutes and check for leaks.

Marine Dual/Triple Filter Application

- The Sea Pro™ filter head **must** be checked for leaks.
- Check the plumbing to and from the Stage 1 and Stage 2 filter heads.
- Reset the Delta P indicator by firmly pressing down on the pop-up indicator.
- Operate the engine for 1 to 2 minutes and check for leaks.



Marine Triplex/Quadplex Application

- Check the Sea Pro™ filter head for leaks while the engine is running at idle.
- Check the plumbing to and from the Stage 1 and Stage 2 filter heads.
- Reset the Delta P indicator by firmly pressing down on the pop-up indicator.
- Operate the engine for 1 to 2 minutes above idle or reduced capability and check for leaks.

